

Project Report: Automated Employee Onboarding & Offboarding System



Project Title: Automated Employee Onboarding & Offboarding System

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GitHub Repository: <https://github.com/DVINDHU123/Automated-Employee-Onboarding-Off-boarding-System>

1. Abstract:

The Automated Employee Onboarding & Offboarding System is a ServiceNow-based solution designed to automate the complete lifecycle of employee onboarding and offboarding requests. The system enables HR teams to raise requests through a Service Catalog interface, which are then processed automatically using Flow Designer.

The workflow includes request creation, manager approval handling, automated departmental task assignments for IT, Facilities, and Security teams, email notifications, and lifecycle record updates. This system minimizes manual HR intervention, ensures faster processing, enforces role-based access control, and enhances transparency in employee lifecycle management

2. Problem Statement:

In traditional HR environments, handling employee onboarding and offboarding involves manual processes like emails, spreadsheets, and physical paperwork. These approaches lead to:

- Delays in account creation, asset provisioning, and workspace setup
- Lack of centralized tracking of onboarding/offboarding status
- Human errors in task assignment and data entry
- Poor communication between HR, IT, Facilities, and Security teams
- No audit trail for approvals and task completions

Hence, there is a need for an automated system that efficiently handles the complete employee lifecycle with proper tracking, approvals, and notifications.

3. Proposed Solution:

The proposed system uses ServiceNow Service Catalog and Flow Designer to:

- Provide a user-friendly request interface via Service Portal
- Automate backend processing using Flow Designer workflows
- Implement manager approval mechanisms with email notifications
- Auto-assign departmental tasks to IT, Facilities, and Security teams
- Maintain centralized records in the Employee Lifecycle table
- Enforce role-based access control using ACLs
- Track SLA compliance for all departmental tasks

4. Scope of the Project:

This system is applicable in:

- HR Service Management for employee lifecycle automation
- IT Service Management (ITSM) for asset and account provisioning
- Corporate IT support teams managing new hire and exit processes
- Multi-department coordination (HR, IT, Facilities, Security)
- Organizations looking to digitize and automate manual HR workflows

5. Methodology:

The project follows a step-by-step approach:

Step 1: Service Catalog Item Creation

A catalog item titled Onboard/Offboard Employee was created to allow HR teams to submit onboarding and offboarding requests..

Catalog UI Policy

Show Exit Date for Offboarding

UpdateDelete

Catalog UI policies are similar to standard UI policies. Catalog UI policies dynamically change variables that are part of a catalog item or change how variable sets are handled. Policies can also be applied when the variables are present in a Requested Item or Catalog Task form. [More Info](#)

Applies to

A Catalog Item

Application

Global

* Catalog Item

Onboard/Offboard Employee

Active

* Short description

Show Exit Date for Offboarding

When to Apply

Script

Catalog UI policy actions are applied only if all the following conditions are met:

1. The catalog UI policy is Active

2. The items in the Conditions field evaluate to true

3. The field specified in the catalog UI policy is present on the specified catalog item

Catalog Conditions

Add Filter Condition

Add OR Clause

request_type

is

Offboarding

AND

OR

Applies on a Catalog Item view

Applies on Catalog Tasks

Applies on Requested Items

Apply the catalog UI policy actions when the form is loaded or when the user changes values on the form

On load

Reverse the effects of the catalog UI policy actions when the Conditions evaluate to false

Reverse if false

Update

Delete

Related Links

[Run Point Scan](#)

[\[SN UI\] Versions \(1\)](#)

Catalog UI Policy Actions

Order

Search

Actions on selected rows...

New

UI policy = Show Exit Date for Offboarding

Name	Read only	Mandatory	Visible	Order
exit_date	Leave alone	Leave alone	True	100

1 to 1 of 1

Step 2: Defining Variables and Variable Sets

A **Variable Set** was configured to capture inputs like:

A Variable Set named Employee Core Details was configured to capture inputs like Request Type, Employee ID, Department, Joining/Exit Date, Manager, and Assets Required.

servicenow

All

Favorites

History

Admin

Variable Set - Requestor Information

Search

Main

Schedules

Maintenance Schedules

Service Catalog

Catalog Definitions

Maintain Catalogs

Maintain Categories

Maintain Dynamic Categories

Maintain Items

Maintain Cart Layouts

Catalog Administration

Request Parent Mapping

Service Catalog Wizards

Maintain Wizards

Contract

Contracts

Maintenance

Variable Set

Requestor Information

Update

Delete

Title

Requestor information

Application

Global

Internal name

requestor_information

Display title

Order

200

Layout

2 Columns Wide, one side, then 1

Type

Single Row

Description

Update

Delete

Variables (5)

Catalog UI Policies

Catalog Client Scripts

Included In (1)

Catalog Data Lookup Definitions

Order

Search

Actions on selected rows...

New

Variable set - Requestor Information

Name	Type	Question	Order
opened_on_behalf_of	Reference	Opened on behalf of	100
email_id	Single Line Text	Email Id	200

Variable Set

Employee Core Details

Update

Delete

Title

Employee Core Details

Application

Global

Internal name

employee_core_details

Variable Set attributes

Order

100

Layout

1 Column Wide

Type

Multi Row

Description

Update

Delete

SN Utilis Versions (15)

Variables (10)			
Catalog UI Policies			
Catalog Client Scripts			
Included In			
Catalog Data Lookup Definitions			
Type			
Search			
Actions on selected rows...			
New			
Variable set - Employee Core Details			
Name	Type	Question	Order
access_revocation_required	Yes / No	Access Revocation Required	
comments	Multi Line Text	Comments	
assets_required	Multiple Choice	Assets Required	
request_type	Select Box	Request Type	
employee_id	Single Line Text	Employee ID	
department	Reference	Department	
employee_name	Reference	Employee Name	
manager	Reference	Manager	
joining_date	Date	Joining Date	
exit_date	Date	Exit Date	
1 to 10 of 10			

Step 3: Flow Designer – Initial Setup

A new flow named On Or Off Boarding Flow was created in Flow Designer triggered when the catalog item is submitted

The screenshot shows the Workflow Studio interface for a flow named "on or off Boarding Flow". The flow is currently "Inactive". The trigger is "Service Catalog". The actions are as follows:

1. Create Employee Lifecycle Record
2. Ask For Approval on Requested Item
3. Create Catalog Task Record
4. Create Catalog Task Record
5. Create Catalog Task Record
6. Wait For Catalog Task Condition where (Request item is [Trigger -> Requested Item ...], and State is Closed Complete)
7. Update Employee Lifecycle Record
8. Update Requested Item Record
9. Send Email

The right-hand pane shows the "Data" section with the following variables:

- Flow Variables
 - Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Table Name (Table Name)
 - Run Start Date/Time (Date/Time)
- 1 - Create Record
 - Employee Lifecycle Record (Record)
 - Employee Lifecycle Table (Table)
 - Action Status (Object)
- 2 - Ask For Approval
 - Approval State (Choice)
 - Action Status (Object)
- 3 - Create Record
 - Catalog Task Record (Record)
 - Catalog Task Table (Table)
 - Action Status (Object)
- 4 - Create Record
 - Catalog Task Record (Record)
 - Catalog Task Table (Table)

Step 4: Get Catalog Variables Action

This step fetches the submitted variables for further processing.

The screenshot shows the Workflow Studio interface for the same flow, now "Active". The trigger remains "Service Catalog". The actions are updated:

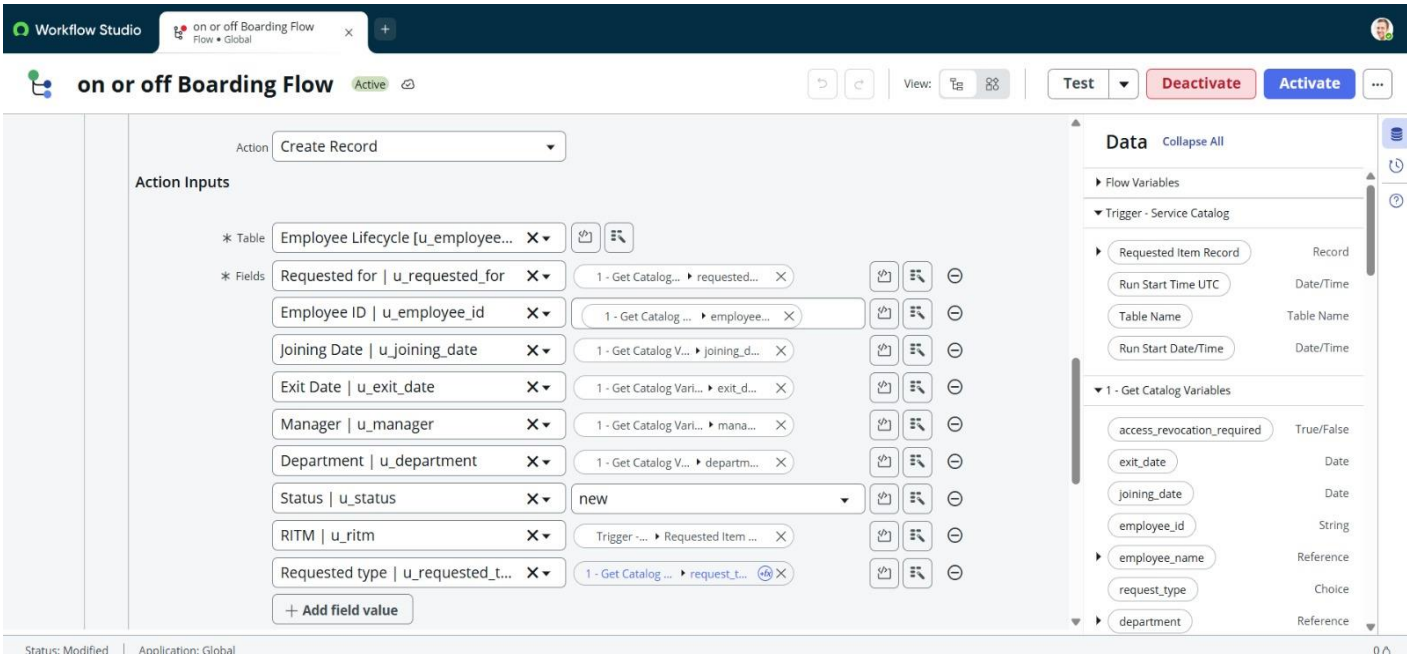
1. Get Catalog Variables from Onboard/Offboard Employee
2. Ask For Approval on Requested Item

The right-hand pane shows the "Data" section with the following variables:

- Flow Variables
 - Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Table Name (Table Name)
 - Run Start Date/Time (Date/Time)

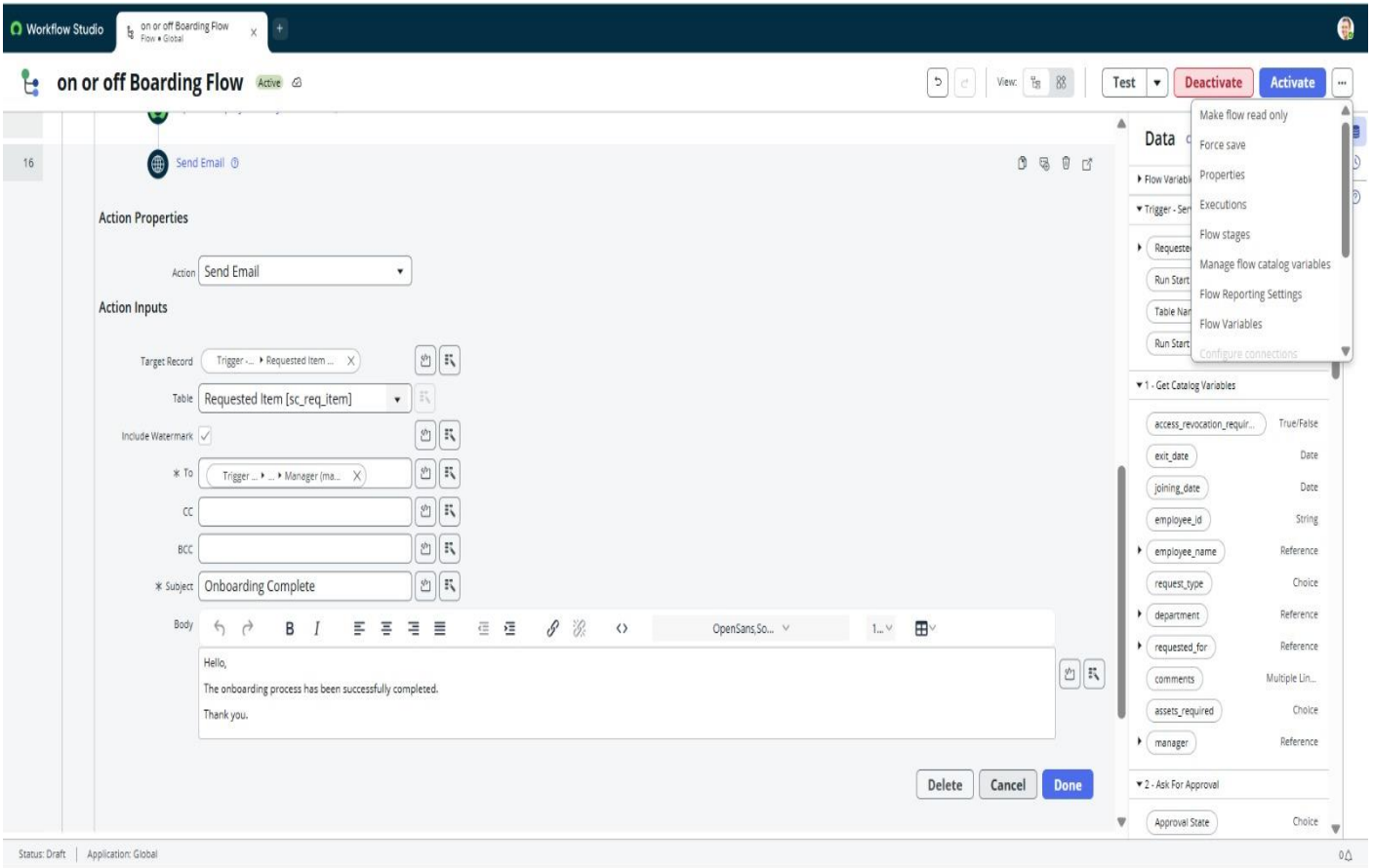
Step 5: Create Record Action

Automatically creates a record in the Employee Lifecycle table for tracking.



Step 6: Send Email Action

An automated email notification is sent to the requester confirming the request submission.



Step 7: Ask for Approval Action

The request is routed to the Manager for authorization before departmental tasks are create

The screenshot displays the Workflow Studio interface for configuring the 'on or off Boarding Flow'. The main workspace shows two steps: 1. 'Get Catalog Variables from Onboard/Offboard Employee' and 2. 'Ask For Approval on Requested Item'. The 'Ask For Approval' action is selected, and its properties are visible on the left.

Action Properties:

- Action: Ask For Approval

Action Inputs:

- * Record: Trigger -> Requested item ...
- Table: Requested Item [sc_req_item]
- Approval Reason: (empty)
- Approval Field: Approval
- Journal Field: Approval history
- * Rules: Approve When: Anyone approves 1 - Get Catalog Vari... -> mana...
- Due Date: None

Data Panel (Right):

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record: Record
 - Run Start Time UTC: DateTime
 - Table Name: Table Name
 - Run Start Date/Time: DateTime
- 1 - Get Catalog Variables
 - access_revocation_requir...: True/False
 - exit_date: Date
 - joining_date: Date
 - employee_id: String
 - employee_name: Reference
 - request_type: Choice
 - department: Reference
 - requested_for: Reference
 - comments: Multiple Lin...
 - assets_required: Choice
 - manager: Reference
- 2 - Ask For Approval
 - Approval State: Choice

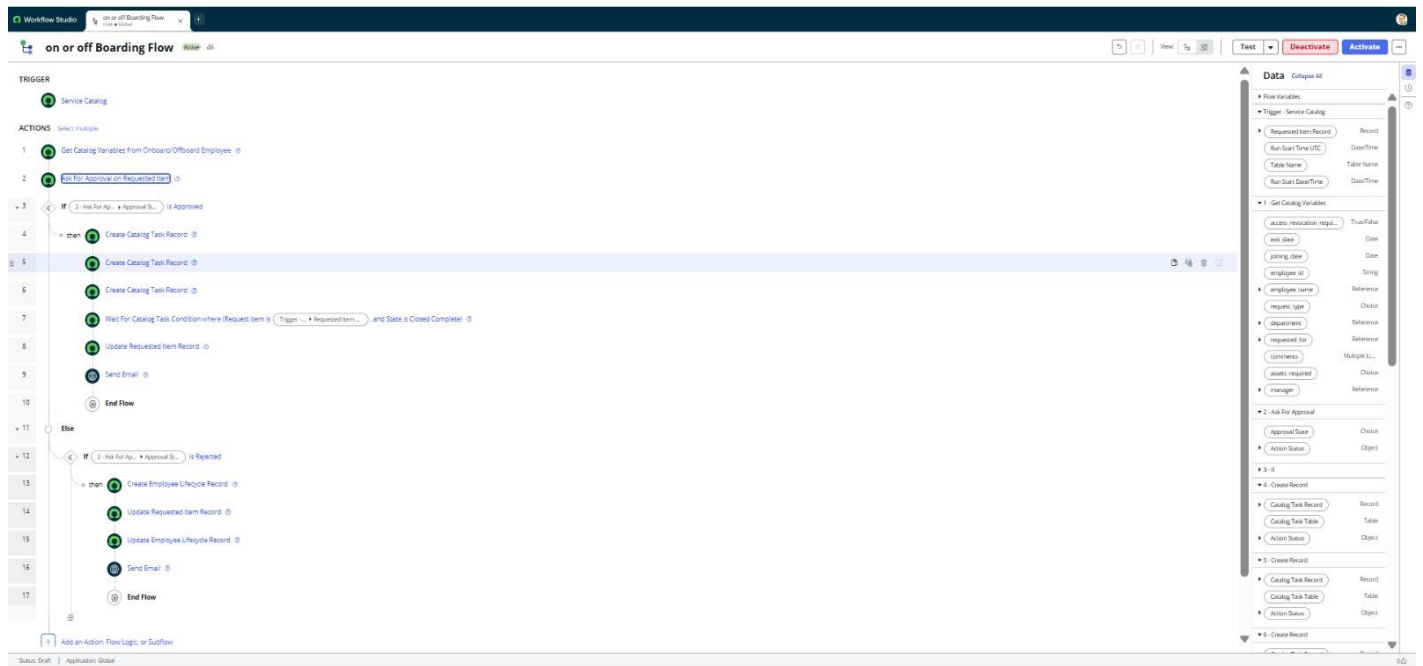
Buttons: Test, Deactivate, Activate, Add another OR rule set, Delete, Cancel, Done.

Status Bar: Status: Draft | Application: Global

Step 8: Conditional Flow Logic

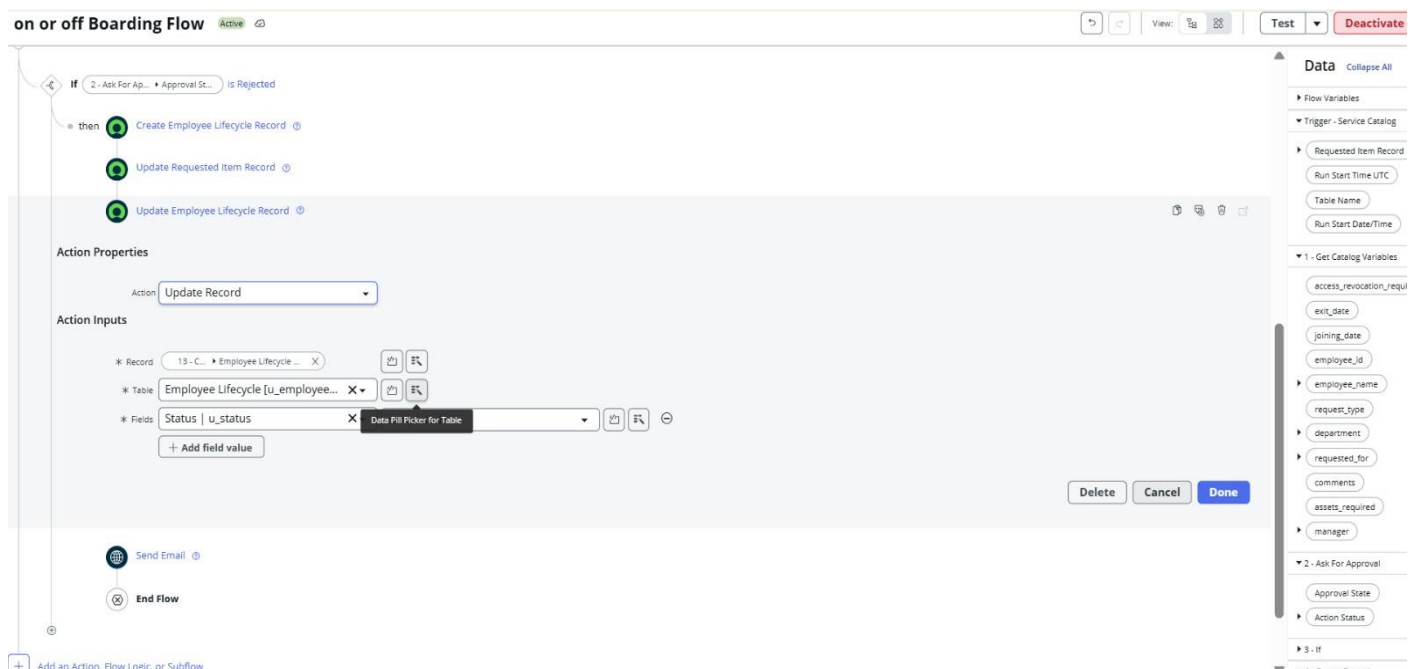
Based on approval:

1. **If Approved** → create departmental tasks for IT, Facilities, and Security.
2. **If Rejected** → Notify requester with reason.



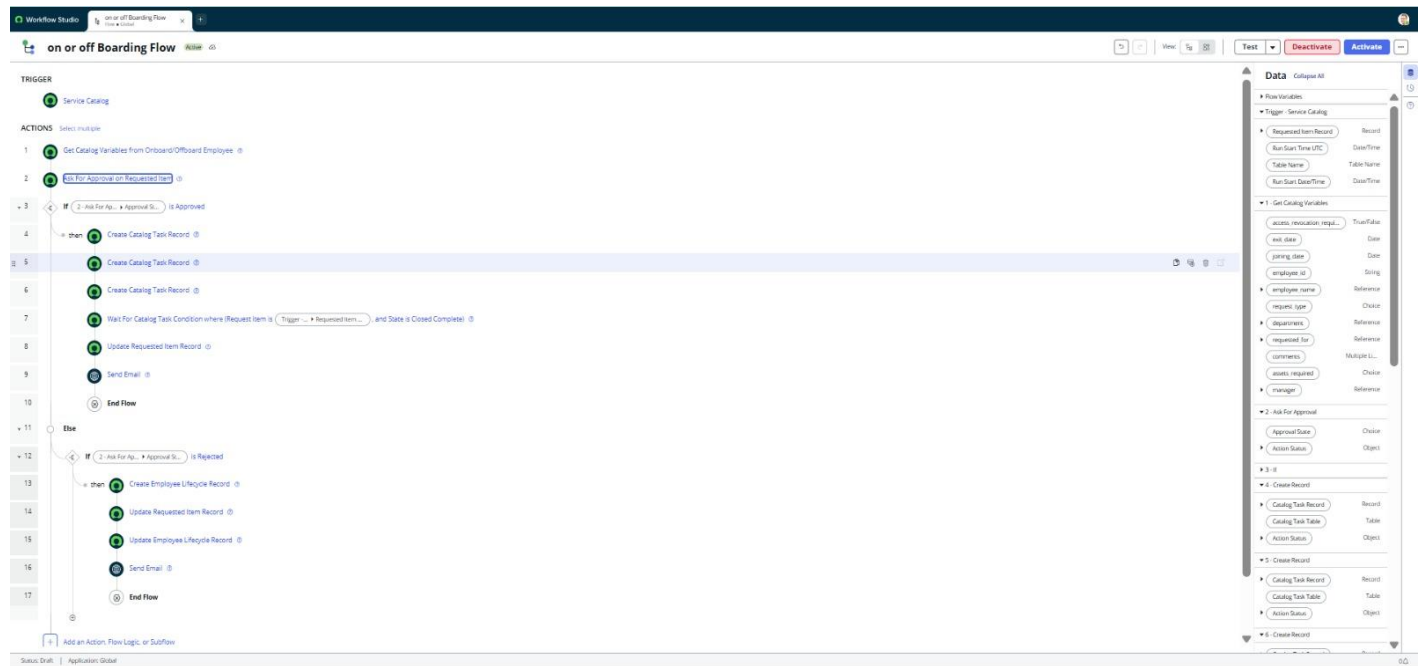
Step 9: Update Record Action

The Employee Lifecycle record and RITM are updated with the latest status (Completed/Rejected).



Step 10: Final Flow Design

The complete workflow is now automated end-to-end.



6.Results

- HR teams can now raise onboarding and offboarding requests directly from the Service Portal.
- Each request is automatically logged, tracked, and routed to the Employee Lifecycle table.
- Approvals and notifications are fully automated through Flow Designer.
- Departmental tasks are auto-assigned to IT, Facilities, and Security teams upon approval.
- SLA tracking ensures all tasks are completed before the employee joining or exit date.
- Role-based ACLs ensure each team accesses only their relevant tasks and data.
- Complete audit trail is maintained for all approvals and task completion

7.Conclusion:

The Automated Employee Onboarding & Offboarding System effectively demonstrates how ServiceNow can be used to automate complex HR service workflows. Built entirely on the ServiceNow platform using Service Catalog, Flow Designer, and Role-Based ACLs, the system significantly improves operational efficiency, enhances user experience, and provides structured and traceable request handling.

The system eliminates manual coordination between HR, IT, Facilities, and Security teams by automating task creation, approval routing, and email notifications. The Employee Lifecycle table provides complete visibility into the status of each onboarding and offboarding request, enabling data-driven HR decisions.

8.Future Enhancements:

- AI-based issue classification
- Chatbot integration
- SLA tracking system
- Dashboard analytics
- Mobile app integration

References

1. ServiceNow Documentation- <https://docs.servicenow.com/>
2. GitHub Project Repository- <https://github.com/DVINDHU123/Automated-Employee-Onboarding-Offboarding-System>

